E-Commerce US Dataset Analysis

Report 5: Exploratory Data Analysis (EDA)

Nithish | AI/ML Engineer

1. Analysis Scope

- Source: master_clean.csv (10,000 orders × 39 columns) and items_clean.csv (21,800 rows).
- 15 visualizations produced and saved under reports/EDA visuals/.
- Analysis split across six dimensions: customers, orders, products, sellers, payments, and reviews.

2. Customer Analysis

2.1 Geographic Distribution

- CA leads with 25.5% of orders (2,546 orders), followed by TX at 18.9% (1,885) and FL at 13.1% (1,312).
- Top 3 states account for 57% of total volume.
- Remaining 7 states share 43%, with CO, WA, and AZ each contributing under 3%.

2.2 Demographics

- Gender split: 64.3% Female, 35.7% Male.
- Age distribution (post-treatment): mean 44.5 years, range 19–70.
- Age peaks in the 35–55 bracket — core buying demographic.

2.3 Customer Segments

Segment	Order Share	Characteristics
Consumer	69.8%	Individual buyers; highest volume
Corporate	20.5%	Business accounts; larger basket sizes
VIP	9.8%	High-spend repeat customers; targeted for loyalty programs

2.4 Repeat Purchase Rate

- 10,000 orders placed by 7,898 unique customers → approximately 21% repeat rate.
- Monthly customer acquisition trend shows a steady inflow with seasonal peaks.

3. Order Performance

3.1 Order Status

- 9,332 delivered (93.3%), 668 canceled (6.7%).
- Cancellation rate of 6.7% is a baseline for monitoring fulfillment risk.

3.2 Delivery Performance

- 90.4% of delivered orders arrived on or before the estimated delivery date.
- Average actual delivery time: 7.5 days.
- 9.6% late deliveries (arrived after estimated date) represent operational improvement opportunity.

3.3 Temporal Patterns

- Order volume shows year-over-year monthly consistency with seasonal demand peaks.
- End-of-week orders (Friday–Sunday) run higher than weekday volume.
- No single month dominates; demand is broadly distributed across the year.

4. Product Analysis

4.1 Category Performance

Category	Item Count	Share of Orders
Electronics	3,860	38.6%

Fashion	2,002	20.0%
Books	1,451	14.5%
Home Goods	~800	~8%
Auto	~700	~7%
Furniture	~650	~6%
Toys	~600	~6%

- Electronics accounts for more than a third of all items sold — single largest category.
- Fashion and Books together cover another 34.5%, making three categories the primary revenue drivers.

4.2 Pricing

- Product price distribution is right-skewed; bulk of the catalog sits in the $100–$500 range.
- 4% of item prices exceed $1,746 — a valid premium tier rather than data errors.

5. Seller Ecosystem

5.1 Revenue Concentration

- 362 sellers (72% of the 500-seller base) generate 80% of total revenue — a classic Pareto split.
- Top seller (Thomas and Sons) significantly outpaces all others.
- Bottom 28% of sellers (138 sellers) share only 20% of revenue.

5.2 Geographic Distribution

- Seller locations mirror customer locations: CA, TX, and NY lead.
- Demand-supply alignment is strong — top buying states also host the top selling companies.

6. Payment Analysis

6.1 Payment Method Breakdown

Payment Method	Order Share
Credit Card	50.3%
PayPal	13.0%
Bank Transfer	9.8%
Debit Card	8.9%
Apple Pay	8.6%
Voucher	6.8%
Boleto	2.5%

6.2 Installment Usage

- 46% of orders paid in full (single installment).
- 54% of orders paid in installments (2–12 payments), primarily via credit card.
- Deferred payment options appear to be a deciding factor for completing higher-value orders.

6.3 Revenue

- Total platform revenue from the 10,000-order sample: approximately $12.09 million.
- Average order value (AOV): approximately $1,209.
- Monthly revenue tracks monthly order volume closely — spend per order is consistent over time.

7. Customer Satisfaction

7.1 Review Scores

- Average review score: 4.05 / 5.0 across 9,332 reviewed orders.
- Score distribution skews toward 4–5 stars; 1–2 star reviews are a minority but warrant attention.

7.2 Satisfaction vs. Delivery Speed

Delivery Speed	Delivery Window	Avg Review Score
Fast	< 5 days	~4.3
Medium	5–10 days	~4.0
Slow	> 10 days	~3.6

- Faster delivery correlates directly with higher satisfaction scores.
- Logistics speed is the primary lever for improving customer ratings.

8. Key EDA Takeaways

- Geographic concentration in CA/TX/FL creates both scale advantage and dependency risk.
- Electronics dominance (38.6%) leaves the platform exposed to a single-category slowdown.
- Strong delivery execution (90.4% on-time) is a competitive asset worth maintaining.
- Installment usage (54%) signals price sensitivity — removing financing options would likely reduce conversion.
- 21% repeat rate is a starting point; the VIP segment (9.8%) is the retention anchor to grow.